

Dạng toàn phương về chính tắc

I. DẠNG TOÀN PHƯƠNG

Trực giao

Lagrange: nhóm các số hạng

Jacobi

$$A \rightarrow \lambda \rightarrow P \rightarrow P_e \rightarrow D \rightarrow \begin{cases} f(Y) = Y^T D Y, \\ X = P_e Y \rightarrow Y = P_e^{-1} X = P_e^T X \end{cases}$$

$$A \rightarrow \begin{cases} \Delta_1 \\ \Delta_2 \\ \Delta_3 \\ \vdots \\ \Delta_n \end{cases} \rightarrow \begin{cases} f(Y) = \frac{1}{\Delta_1} y_1^2 + \frac{\Delta_2}{\Delta_1} y_2^2 + \frac{\Delta_3}{\Delta_2} y_3^2 + \dots + \frac{\Delta_n}{\Delta_{n-1}} y_n^2 \\ X = PY \leftarrow P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & \cdot & p_{1n} \\ 0 & p_{22} & p_{23} & \cdot & \cdot \\ 0 & 0 & p_{33} & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & 0 & \cdot & p_{nn} \end{pmatrix} \leftarrow \begin{pmatrix} a_{11} & \cdot & \cdot & \cdot & a_{1n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & \cdot & \cdot & \cdot & a_{nn} \end{pmatrix} \begin{pmatrix} p_{1i} \\ p_{2i} \\ \cdot \\ \cdot \\ p_{ni} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 1 \end{pmatrix} \end{cases}$$

VD1:

Phép biến đổi trực giao

$$f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1 \longrightarrow A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} \longrightarrow \begin{cases} \lambda = 4 \rightarrow u_1 = (1, 1, 1) \\ \lambda = 1 \rightarrow \begin{cases} u_2 = (1, -1, 0) \\ u_3 = (1, 1, -2) \end{cases} \end{cases} \longrightarrow P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 0 & -2 \end{pmatrix}$$

$$D = P_e^{-1} A P_e = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \longleftarrow P_e^{-1} = P_e^T = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{-2}{\sqrt{6}} \end{pmatrix} \longleftarrow P_e = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{pmatrix}$$

$$f(y_1, y_2, y_3) = Y^T D Y = 4y_1^2 + y_2^2 + y_3^2, Y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$X = P_e Y \rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{-2}{\sqrt{6}} \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \rightarrow \beta = \left\{ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right), \left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, 0 \right), \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \right) \right\}$$

Phép biến đổi Jacobi

VD2:

$$f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_2x_3 + 2x_3x_1$$

$$\Rightarrow A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$\Delta_1 = \det(2) = 2, \quad \Delta_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3, \quad \Delta_3 = \det \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 2 & 2 \end{pmatrix} = 4$$

$$\downarrow$$

$$f(y_1, y_2, y_3) = \frac{1}{2}y_1^2 + \frac{3}{2}y_2^2 + \frac{4}{3}y_3^2$$

$$* a_{11}p_{11} = 1 \rightarrow 2p_{11} = 1 \rightarrow p_{11} = 1/2$$

$$* \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} p_{12} \\ p_{22} \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \rightarrow \begin{cases} p_{12} = -1/3 \\ p_{22} = 2/3 \end{cases}$$

$$\nearrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 2 & -1/2 & -1/4 \\ 0 & 2/3 & -1/4 \\ 0 & 0 & 3/4 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$$

$$* \begin{pmatrix} a_{11} & a_{11} & a_{11} \\ a_{11} & a_{11} & a_{11} \\ a_{11} & a_{11} & a_{11} \end{pmatrix} \begin{pmatrix} p_{13} \\ p_{23} \\ p_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} p_{13} \\ p_{23} \\ p_{33} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \rightarrow \begin{cases} p_{13} = -1/4 \\ p_{23} = -1/4 \\ p_{33} = 3/4 \end{cases}$$

$$\downarrow$$

$$\beta = \left\{ (2, 0, 0), \left(\frac{-1}{2}, \frac{2}{3}, 0 \right), \left(\frac{-1}{4}, \frac{-1}{4}, \frac{3}{4} \right) \right\}$$

Phép biến đổi Lagrange

VD3:

$$\begin{aligned}f(x_1, x_2, x_3) &= 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3 \\&= 2 \left[x_1^2 + 2x_1 \cdot \frac{x_2 + x_3}{2} + \left(\frac{x_2 + x_3}{2} \right)^2 \right] - \frac{1}{2}(x_2 + x_3)^2 + 2x_2^2 + 2x_3^2 + 2x_2x_3 \\&= 2 \left(x_1 + \frac{x_2}{2} + \frac{x_3}{2} \right)^2 + \frac{3}{2}x_2^2 + \frac{3}{2}x_3^2 + x_2x_3 \\&= 2 \left(x_1 + \frac{x_2}{2} + \frac{x_3}{2} \right)^2 + \frac{3}{2} \left(x_2^2 + 2x_2 \cdot \frac{x_3}{3} + \frac{x_3^2}{9} \right) - \frac{1}{6}x_3^2 + \frac{3}{2}x_3^2 \\&= 2 \left(x_1 + \frac{x_2}{2} + \frac{x_3}{2} \right)^2 + \frac{3}{2} \left(x_2 + \frac{x_3}{3} \right)^2 + \frac{4}{3}x_3^2\end{aligned}$$

$$f(y_1, y_2, y_3) = 2y_1^2 + \frac{3}{2}y_2^2 + \frac{4}{3}y_3^2$$

$$\begin{cases} y_1 = x_1 + \frac{x_2}{2} + \frac{x_3}{2} \\ y_2 = x_2 + \frac{x_3}{3} \\ y_3 = x_3 \end{cases}$$



$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 1 & 1/2 & 1/2 \\ 0 & 1 & 1/3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$



$$X = PY$$



β

VD4: đưa f về
dạng toàn phương
chính tắc

$$1) f(x_1, x_2) = 5x_1^2 + 8x_2^2 - 4x_1x_2$$

$$2) f(x_1, x_2, x_3) = x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3$$

$$4) f(x_1, x_2, x_3) = 4x_1^2 + 4x_2^2 + 4x_3^2 + 4x_1x_2 + 4x_1x_3 + 4x_2x_3$$

$$5) f(x_1, x_2, x_3) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3$$

$$6) f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_3 - 2x_2x_3$$

$$7) f(x_1, x_2, x_3) = -2x_1^2 - 5x_2^2 - 5x_3^2 - 4x_1x_2 + 4x_1x_3 + 8x_2x_3$$

$$8) f(x_1, x_2, x_3) = 2x_1^2 + 5x_2^2 + 5x_3^2 + 4x_1x_2 - 4x_1x_3 - 8x_2x_3$$

$$9) f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 5x_3^2 + 2x_1x_2 + 4x_1x_3 + 4x_2x_3$$

$$10) f(x_1, x_2, x_3) = -x_1^2 - x_2^2 + 2x_1x_2 - 6x_1x_3 + 8x_2x_3$$

VD5: đưa f về dạng toàn phương chính tắc

$$1) \quad f(x_1, x_2, x_3) = 4x_1x_2 + 4x_1x_3 + 4x_2x_3$$

$$2) \quad f(x_1, x_2, x_3) = x_1x_2 + 4x_1x_3 + x_2x_3$$

$$3) \quad f(x_1, x_2, x_3) = 2x_1x_2 - 4x_1x_3 + 6x_2x_3$$

II. VECTOR RIÊNG, TRỊ RIÊNG, CHÉO HÓA MA TRẬN

VD6: (a) $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} \Rightarrow ? : A^{10} \text{ \& } A^n$

(b) $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \\ 2 & -2 & 3 \end{bmatrix} \Rightarrow ? : A^{10} \text{ \& } A^n$

(c) Cho ma trận thực $A = \begin{pmatrix} 5 & 6 \\ -2 & -3 \end{pmatrix}$.

Hãy chéo hóa A , rồi sau đó tìm A^m , với mọi m nguyên; $m \geq 0$.

(d) Cho ma trận thực $A = \begin{pmatrix} 7 & 9 \\ -2 & -4 \end{pmatrix}$.

Hãy chéo hóa A , rồi sau đó tìm A^m , với mọi m nguyên; $m \geq 0$.